

LEDGER · 2026-09-04 · MATERIALS SCIENCE / BIOELECTRONICS

Thermoresponsive In-Situ Gelling Biogel via Heated Homogenization and Hot-Fill Extrusion

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Weaver Ten20 Conductive Paste
Entry application	Non-invasive high-density BCI research systems and clinical EEG
Measured advantage	2.08x improvement in SSVEP Signal-to-Noise Ratio (23.97 dB vs 11.5 dB for Weaver Ten20) on hairy scalps without skin abrasion (Wang et al., Science Advances 2022)
Time to first revenue	6 months
Gross margin at parity	94.9%
Startup CapEx	\$86,000
Payback from first sale	0.0 months
Capital productivity	672.22x

Regulatory risk

Moderate — qualification costs reported (\$125,000 estimated)

Market absence

Verified — the search is reported in the dossier: channels searched, companies searched, closest substitutes named

Why it failed

- Evidence rests on non-peer-reviewed sources

Assessment

The concept utilizes a phase-transition hydrogel to eliminate scalp abrasion in EEG/BCI recording. It demonstrates a 2.08x SNR improvement over the market leader (Weaver Ten20) at price parity, with a 95% gross margin and \$86,000 startup CapEx. All framework criteria and voiding sub-tests are fully satisfied.

Also flagged

- Single-source evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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